

# HASHING

## PROBLEM 3

### QUESTION:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |         |                      |         |              |
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| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Session | Hashing | Question Information | Level 1 | Challenge 93 |
| <p><b>Question description</b></p> <p>Alice likes festivals a lot as we all do. He also likes spending money on these festivals. He spends money in buying various things on these festivals. But he has problem of forgetting. He only remembers his top three maximum spendings for any festival.</p> <p>For eg, on Holi he spends 25 units on colors, 50 units on water sprays, 100 units on gifts, 150 units on sweets but he remebers only his top 3 spendings ,i.e., 50, 100 &amp; 150.</p> <p>Now as the year ends he wants to know the <b>festival on which he spent most of his money that he can remember</b>.</p> <p><b>Constraints</b></p> $1 \leq T \leq 100$ $1 \leq N \leq 10^4$ $1 \leq X \leq 10^9$ $1 \leq  S  \leq 10;  S  \text{ denotes string length}$ <p>S will consist of <b>lowercase and/or uppercase English aplhabets</b></p> <p><b>Input Format</b></p> <p>First line contains an integer T, denoting number of test cases.</p> <p>Second line contains an integer N, denoting number of spendings in a year.</p> <p>Each of the next N lines contain a string S and an integer X denoting festival name and spending on one of the events of that festival.</p> <p><b>Output Format</b></p> <p>For each test case, print a single line containing the festival name and its total spending that he remembers.</p> <p>If there are more than 1 festival with the maximum spending, print the one which has lexicographically smallest name.</p> |         |         |                      |         |              |

Logical Test Cases

|                                                                                                                                                                                                 |                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Test Case 1</p> <p>INPUT (STDIN)</p> <p>2<br/>6<br/>B 20<br/>A 2<br/>A 10<br/>A 10<br/>B 30<br/>A 30<br/>3<br/>abc 10<br/>xyz 15<br/>oop 8</p> <p>EXPECTED OUTPUT</p> <p>A 50<br/>xyz 15</p> | <p>Test Case 2</p> <p>INPUT (STDIN)</p> <p>2<br/>5<br/>BA 25<br/>AD 24<br/>Ag 10<br/>AE 15<br/>AT 30<br/>3<br/>abc 15<br/>xyz 25<br/>oops 8</p> <p>EXPECTED OUTPUT</p> <p>AT 30<br/>xyz 25</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Mandatory Test Cases

|                                                     |
|-----------------------------------------------------|
| <p>Test Case 1</p> <p>KEYWORD</p> <p>while(t--)</p> |
|-----------------------------------------------------|

Complexity Test Cases

|                                                          |                                                  |                                          |
|----------------------------------------------------------|--------------------------------------------------|------------------------------------------|
| <p>Test Case 1</p> <p>CYCLOMATIC COMPLEXITY</p> <p>7</p> | <p>Test Case 2</p> <p>TOKEN COUNT</p> <p>260</p> | <p>Test Case 3</p> <p>NLOC</p> <p>46</p> |
|----------------------------------------------------------|--------------------------------------------------|------------------------------------------|

## SOURCE CODE:

```
#include <stdio.h>
#include <string.h>

typedef struct EventRecord {
    char title[20];
    long long top_bills[3];
} EventRecord;

EventRecord records[1005];
int recorded_events = 0;

int locate_or_register(const char* name) {
    for (int i = 0; i < recorded_events; i++) {
        if (strcmp(records[i].title, name) == 0) return i;
    }
    strcpy(records[recorded_events].title, name);
    records[recorded_events].top_bills[0] = 0;
    records[recorded_events].top_bills[1] = 0;
    records[recorded_events].top_bills[2] = 0;
    return recorded_events++;
}

void register_expense(int idx, long long cost) {
    if (cost > records[idx].top_bills[0]) {
        records[idx].top_bills[2] = records[idx].top_bills[1];
        records[idx].top_bills[1] = records[idx].top_bills[0];
        records[idx].top_bills[0] = cost;
    } else if (cost > records[idx].top_bills[1]) {
        records[idx].top_bills[2] = records[idx].top_bills[1];
        records[idx].top_bills[1] = cost;
    } else if (cost > records[idx].top_bills[2]) {
        records[idx].top_bills[2] = cost;
    }
}

int main() {
    int test_cases;
    if (scanf("%d", &test_cases) != 1) return 0;

    while (test_cases--) {
        int entries;
        scanf("%d", &entries);
        recorded_events = 0;

        for (int i = 0; i < entries; i++) {
            char name_buf[20];
            long long amount;
            scanf("%s %lld", name_buf, &amount);
            int event_idx = locate_or_register(name_buf);
            register_expense(event_idx, amount);
        }

        int winner_idx = 0;
        long long peak_expenditure = -1;
        for (int i = 0; i < recorded_events; i++) {
            long long total = records[i].top_bills[0] + records[i].top_bills[1] + records[i].top_bills[2];
            if (total > peak_expenditure ||
                (total == peak_expenditure && strcmp(records[i].title, records[winner_idx].title) < 0)) {
                peak_expenditure = total;
                winner_idx = i;
            }
        }
        printf("%s %lld\n", records[winner_idx].title, peak_expenditure);
    }
    return 0;
}
```

## OUTPUT:

```
1
6
B 20
BA 25
AD 24
A 10
A 10
A 30
A 50
```

```
2
5
BA 25
AD 24
Ag 10
AE 15
AT 30
3
abc 15
xyz 25
oops 8
AT 30
xyz 25
```